

Example: Calculation of Covariance

- Suppose two stocks X_1 and X_2 have the following values in one week:
 - $(2, 5), (3, 8), (5, 10), (4, 11), (6, 14)$
- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?

- Covariance formula

$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E[X_1 X_2] - \mu_1 \mu_2 = E[X_1 X_2] - E[X_1]E[X_2]$$

- Its computation can be simplified as: $\sigma_{12} = E[X_1 X_2] - E[X_1]E[X_2]$

- $E(X_1) = (2 + 3 + 5 + 4 + 6) / 5 = 20 / 5 = 4$

- $E(X_2) = (5 + 8 + 10 + 11 + 14) / 5 = 48 / 5 = 9.6$

- $\sigma_{12} = (2 \times 5 + 3 \times 8 + 5 \times 10 + 4 \times 11 + 6 \times 14) / 5 - 4 \times 9.6 = 4$

- Thus, X_1 and X_2 rise together since $\sigma_{12} > 0$